Inference as Consolidation: Replay in the Silent Degrees of Freedom

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Abstract

Biological brains consolidate memories both in dedicated offline states (sleep) and, as recent causal evidence shows, through *local sleep*: brief, use-dependent off-periods of individual cortical circuits during wakefulness. In contrast, replay-based continual learning in artificial neural networks almost universally relies on offline consolidation phases or on interleaving replayed samples with the input stream. We ask whether a network trained end-to-end by local, biologically constrained learning rules can consolidate *during inference*, with no offline phase at all. We introduce (i) a *refractory rotation* rule under which units that have just fired sit out the next competition, (ii) an *exact isolation* scheme that confines replay updates to synapses invisible to the current input under k-winner-take-all (k-WTA) dynamics, with per-synapse optimiser state advanced only inside the mask, and (iii) internal control signals — a unit-level homeostatic pressure that triggers replay bursts, and a relative-novelty (fast-over-slow error) gate that decides when rotation runs. The construction inverts the usual direction of non-interfering continual learning: rather than protecting past tasks while learning the present one, the present computation is held invariant while past memories are consolidated through degrees of freedom the current batch exposes as unused. On class-incremental split-MNIST the resulting system reaches $91.3 \pm 0.3\%$ ($92.0 \pm 0.6\%$ with a stateless optimiser) and no offline phase, exceeding the best offline-night schedule ($90.7 \pm 0.4\%$), at a replay-sample cost of $\approx 1.2\times$ the night's; reversing the protected direction (guarding past memories while learning the present, as in null-space methods) collapses to $81.3 \pm 2.9\%$ under identical machinery; masked replay is insensitive to replay batch size down to eight samples, whereas unmasked replay and offline nights degrade sharply at the same sizes. The i.i.d. cost of rotation (1.3–3.0 points) is eliminated, under masked Adam, by the relative-novelty gate on learnable streams (95.0% vs. 95.0% without rotation); under a stateless optimiser the same gate must instead be committed to sleep-bout-like episodes with a minimum on-duration, which then weakly dominate always-on rotation (92.0/94.2 vs. 92.0/93.3 over six seeds). The price reverses into a +2.5 to +6.5 point *gain* on harder data (split CIFAR-10), where the full ordering of methods transfers. We further show that surprise-triggered waking replay fails at any volume, that the trigger must be homeostatic, and that the residual advantage of offline nights is confined to very small episodic buffers on narrow substrates, and that sleep-anchored synaptic down-selection helps precisely where an offline window exists. We provide exactness proofs for the isolation rule and release all code, checkpoints and diagnostics.

1 Problem Formulation

Setting. Let a stream of labelled examples $(x_t, y_t) \in \mathbb{R}^d \times \{1, \dots, C\}$ be presented in T contiguous tasks $\mathcal{T}_1, \dots, \mathcal{T}_T$, where task $\mathcal{T}_\tau$ contains only classes $\mathcal{C}_\tau \subset \{1, \dots, C\}$, the $\mathcal{C}_\tau$ are disjoint, and no task identity is available at test time (class-incremental learning). A network f_θ with a single shared

C-way readout is trained on the stream and evaluated on all C classes after the final task,

$$A_{\text{seq}}(\theta) \;=\; \Pr_{(x,y)\sim\mathcal{D}_{\text{test}}}\big[\arg\max_c f_\theta(x)_c = y\big], \qquad F \;=\; \frac{1}{T-1}\sum_{\tau=1}^{T-1}\big(a_{\tau,\tau} - a_{T,\tau}\big), \tag{1}$$

where $a_{t,\tau}$ is accuracy on task τ after learning task t and F is forgetting.

Constraints. We restrict the learner to a set $\mathcal{B}$ of cortical constraints: no weight transport (feedback synapses are learned, never copied), errors carried by a bounded burst-like channel, Dale's law, sparse population codes (k-WTA on wide layers), sparse distance-dependent connectivity, and a single forward–feedback sweep per example (no iterative relaxation). An episodic memory ("hippocampus") of at most K raw examples with reservoir writing is permitted; K is a budget, not a free parameter.

Objective. Unlike benchmark-chasing continual learning, we optimise a three-way trade-off

$$\max_\theta \;\big(A_{\text{seq}},\, A_{\text{iid}},\, -R,\, -E\big) \quad \text{subject to} \quad \theta \in \mathcal{B}, \tag{2}$$

where A_{iid} is accuracy of the identical learner on the i.i.d. shuffle of the same data (the *static axis*: a brain-like mechanism must not damage ordinary learning), R is the replay cost in *samples* $\times$ *synapses*, and E an event-driven energy proxy (Section 3). The central question of this paper: can the offline consolidation term be removed entirely — consolidation occurring only *during* inference on the wake stream — without losing A_{seq}, and at what cost in A_{iid} and R?

2 Related Work

Sleep-inspired consolidation in ANNs. The Bazhenov line implements sleep as a global offline phase: networks are converted to spiking dynamics and driven by noise under local Hebbian plasticity, recovering old tasks after each new one (Tadros et al., 2022), after several tasks at once (Bazhenov et al., 2026), in spiking networks with coarse alternation of training and sleep (Golden et al., 2022), and in equilibrium-propagation networks combined with small rehearsal buffers (Kubo et al., 2025). Wake–sleep frameworks with NREM and REM stages (Sorrenti et al., 2024; Robinson et al., 2022) and engineering systems such as SIESTA (Harun et al., 2023) likewise schedule consolidation offline. In all of these, sleep is global, offline, and clocked; none implements consolidation confined to currently unused units during ongoing inference.

Gating, subnetwork and history-dependent competition methods. Context-dependent gating (XdG) activates a fixed random subnetwork per task but requires task identity at train and test time (Masse et al., 2018); learned gates (Tilley et al., 2023), dendritic context vectors from input prototypes (Iyer et al., 2022), task-free sparsification by feedforward and top-down errors (Lässig et al., 2023), and gradient-recovered functional masks (McKee et al., 2026) remove the explicit label but derive the gate from the *input identity*. Use-history-dependent suppression of competition, in itself, has classical precedents: refractory winner-take-all dynamics in physiological models of competitive learning (Kaski and Kohonen, 1994), explicit refractory competitive learning in which a recent winner is temporarily barred from winning (Maeda and Miyajima, 1999), and, recently, spiking continual learners whose firing thresholds rise with each unit's activation history to diversify recruitment across tasks (Shen et al., 2024). Our rotation differs in *purpose*, not primitive: recently wake-active units are benched for one competition specifically to enlarge a dynamically reallocated *replay channel* — rotation manufactures plastic degrees of freedom in which consolidation can proceed concurrently with inference, rather than allocating representational resources across tasks.

Null-space and parameter-isolation methods. A mature line of work constructs non-interfering updates by protecting *previous* tasks: projecting new-task gradients orthogonally to old-task gradient or activation subspaces (Farajtabar et al., 2020; Saha et al., 2021), training in the null space of accumulated feature covariance (Wang et al., 2021), combining k-winner sparsity and heterogeneous dropout with such projections (Abbasi et al., 2022), or freezing pruned-and-frozen subnetworks (Golkar et al., 2019; Mallya and Lazebnik, 2018). Our construction inverts the protected direction: the *ongoing wake computation* is held invariant while *old-memory replay* is written into degrees of freedom that the current batch's exact sparse support exposes as instantaneously invisible — no stored bases, SVDs, or task masks, and the "null space" changes every batch for free. McKee et al. (2026) note that optimiser state and decoupled weight decay can break structural guarantees even at zero gradient, and Bricken et al. (2023) analyse stale momentum in sparse learners; we adopt the corresponding requirement (mask-confined moments) as part of the exactness construction rather than claiming its discovery. Replay *scheduling* has likewise been studied as a learned policy (Klasson et al., 2023); our contribution there is the specific unit-level use-pressure controller and its empirical dissociation from novelty triggering.

Complementary learning systems. CLS-ER maintains fast/slow EMA copies of the weights as semantic memories (Arani et al., 2022) and DualNet trains a slow self-supervised learner beside a fast supervised one (Pham et al., 2021); both raise the value of each replayed sample but still replay at every step. Our contribution is orthogonal: we change *where* (isolated synapses), *when* (homeostatic bursts) and at what granularity (micro-batches) replay is applied.

Pattern separation. Cerebellum-, mushroom-body- and dentate-gyrus-style sparse expansions (Cayco-Gajic et al., 2017; Cayco-Gajic and Silver, 2019; Litwin-Kumar et al., 2017) inspire continual learners such as the FlyModel (Shen et al., 2021), sparse distributed memory (Bricken et al., 2023) and DG-gated mixtures (Kapoor et al., 2026). We tested this family as a front-end and report a negative result (Section 5).

Biology of local sleep. Sleep-like off-periods occur locally in awake, sleep-deprived animals (Vyazovskiy et al., 2011). Causal evidence is recent: optogenetically inducing ON/OFF alternation in one cortical hemisphere of awake mice locally discharges sleep pressure, weakens synaptic markers (GluA1, pS845) and rescues memory consolidation under sleep deprivation, whereas an equal *tonic* reduction of firing does not (Driessen et al., 2026). Our refractory rotation is the algorithmic counterpart of this alternation requirement, and our silent-mask control the counterpart of tonic silencing.

3 Experiment Setup

Data and protocol. Sequential axis: split-MNIST (five tasks of two classes, 5 epochs per task) and split CIFAR-10 (same protocol; 32×32 luminance so that distance-dependent connectivity retains its geometry). Static axis: the identical learner and budgets on the i.i.d. shuffle. Waking batches have $n_w = 16$ samples; the episodic buffer holds $K \in \{200, 1000\}$ raw samples under reservoir writing. All numbers are mean ± s.d. over three seeds unless marked otherwise. A third regime uses a frozen V1-like feature front-end for CIFAR-10: 256 normalised random $6{\times}6{\times}3$ patches sampled from the training images act as fixed filters (stride 2, rectification against the per-filter mean response, quadrant average pooling; 1024-D, linear probe $\approx 49\%$). Nothing in the front-end is learned; the cortex under study receives it as its retina.

Architecture. d–512–256–C with k-WTA sparsity 5% on the hidden layers (d–256–128–C at 10% where noted), 30% distance-dependent connectivity on hidden synapses, Dale's law with 80% excitatory units, and the single-sweep learning rule of Section 4.

Baselines. (i) Backpropagation with experience replay at equal buffer K (mandatory in every table); (ii) the offline *night*: after each waking epoch, 20 replay batches of 256 samples at $3\times$ learning rate with no concurrent input (the strongest schedule from our earlier ablations); (iii) interleaved ER for the local learner; (iv) no buffer.

Metrics. Final accuracy, forgetting F, replay cost R (batches and samples), the isolated-channel width ρ_ℓ (fraction of replay-activated units in layer ℓ that are asleep for the current input), code overlap (mean pairwise Jaccard of task-active unit sets), participation-ratio dimensionality, and an event-driven energy proxy: 0.9 pJ per synaptic event for spiking inference vs. 4.6 pJ per MAC for dense rate inference, with thresholds calibrated post hoc and T_s integration steps (Merolla et al., 2014).

4 Methodology

4.1 Cortical base learner

Layer activities are $a^0 = x$ and, for $\ell = 1, \ldots, L$,

$$z^\ell = g^\ell\left(W^\ell a^{\ell-1}\right) + b^\ell, \qquad a^\ell = \Phi_k\big(\sigma(z^\ell) \odot (1 - s^\ell)\big), \tag{3}$$

where σ is a rectifier, $s^\ell \in \{0,1\}^{n_\ell}$ is the (possibly empty) *suppression mask* of Section 4.3, and Φ_k keeps the k largest entries of each row and zeroes the rest (k-WTA). Errors are produced by one top-down sweep: with target y,

$$\varepsilon^L = y - z^L, \qquad \varepsilon^\ell = \sigma'(z^\ell) \odot \Big(B^\ell\left[\kappa \tanh(\varepsilon^{\ell+1}/\kappa) \odot \mathbf{1}(a^{\ell+1} > 0)\right]\Big), \tag{4}$$

i.e. a bounded, event-gated burst signal carried by learned feedback weights B^ℓ (no transport). Forward and feedback weights follow the same local product with a shared decay λ (Kolen–Pollack alignment (Kolen and Pollack, 1994; Akrout et al., 2019)),

$$\Delta W^\ell \propto \varepsilon^\ell (a^{\ell-1})^\top - \lambda W^\ell, \qquad \Delta B^{\ell-1} \propto \varepsilon^\ell (a^{\ell-1})^\top - \lambda B^{\ell-1}, \tag{5}$$

with sign-concordant feedback under Dale's law and per-synapse adaptive scaling (ϵ-floored Adam). Equations (3)–(5) were fixed by earlier ablation ladders and are not varied here.

4.2 Exactly isolated replay

Let x be the current waking batch and define the *awake set* $\mathcal{A}^\ell(x) = \{i : a_i^\ell(x) \neq 0 \text{ for some sample in } x\}$. During the same forward step, a replay batch r drawn from the buffer is applied through the synapse mask

$$M_{ij}^\ell \;=\; \mathbf{1}\big[\, j \notin \mathcal{A}^{\ell-1}(x) \;\vee\; i \notin \mathcal{A}^\ell(x)\,\big], \tag{6}$$

i.e. a synapse may take the replay update iff its presynaptic unit is silent for the current input or its postsynaptic unit is asleep. We enforce the same constraint on the per-synapse optimiser state (first and second moments advance *only* inside the mask): as McKee et al. (2026) caution and Bricken et al. (2023) analyse for sparse learners, unmasked momentum would otherwise re-inject the isolated update into awake synapses, voiding the exactness guarantees below; the accuracy consequences of this choice are dissected in Section 6. The readout row remains plastic (old classes must stay calibrated).

Proposition 1 (Pre-silent updates are exactly invisible)**.** *Fix input x and let $\widehat{W}^\ell = W^\ell + \Delta^\ell$ with $\Delta^\ell_{ij} = 0$ whenever $a^{\ell-1}_j(x) \neq 0$. Then the network output on x is unchanged for any magnitude of Δ.*

Proof. By induction on ℓ. Assume $a^{\ell-1}$ is unchanged (true for $\ell = 1$ since $a^0 = x$). For any unit i, $\widehat{z}^\ell_i - z^\ell_i = g^\ell \sum_j \Delta^\ell_{ij}\, a^{\ell-1}_j(x)$. Each term vanishes: if $a^{\ell-1}_j(x) \neq 0$ then $\Delta^\ell_{ij} = 0$; otherwise $a^{\ell-1}_j(x) = 0$. Hence z^ℓ, and therefore a^ℓ through (3), is unchanged; the induction closes at the output layer. □

Proposition 2 (Post-asleep updates are invisible under a margin condition)**.** *Let Δ^ℓ additionally allow $\Delta^\ell_{ij} \neq 0$ for units $i \notin \mathcal{A}^\ell(x)$ with active presynaptic partners. Write $\delta_i = g^\ell \sum_j \Delta^\ell_{ij} a^{\ell-1}_j(x)$ and let $z_{(k)}$ be the k-th order statistic of the post-nonlinearity values entering Φ_k. If for every such unit $\sigma(z^\ell_i + \delta_i) < z_{(k)}$ (the perturbed unit stays below the winner threshold), the output on x is unchanged.*

Proof. A unit outside $\mathcal{A}^\ell(x)$ contributes $a^\ell_i(x) = 0$. After the update its pre-activation becomes $z^\ell_i + \delta_i$; under the stated margin it is still excluded by Φ_k (or rectified to zero), so $a^\ell_i(x) = 0$ still, all other units' inputs are untouched by Proposition 1's argument, and the layer's output is identical. □

Corollary 1 (Suppressed units are exact for any magnitude)**.** *If i is suppressed by the rotation mask ($s^\ell_i = 1$ in (3)), then $a^\ell_i(x) = 0$ identically and Proposition 2 holds with no margin condition.*

Remark 1. *Empirically the masked replay update changed the waking batch's output by* 0.0 *(double precision) in all logged checks; Corollary 1 explains why the refractory rotation below makes the isolation robust rather than merely approximate.*

4.3 Refractory rotation and its gates

Definition 1 (Refractory rotation)**.** *After processing waking batch x_t, set $s^\ell_i(t{+}1) = \mathbf{1}[i \in \mathcal{A}^\ell(x_t)]$: every unit that fired sits out the next competition and is therefore asleep — and consolidable — for batch x_{t+1}.*

Rotation widens the isolated channel ρ_ℓ from 0.2–0.3 (natural silence) to ≈ 0.53, at the price of running inference on the second-best coalition. Two internal signals control the system (Figure 1):

Homeostatic pressure (when to replay). Each unit integrates its own use, $S_i \leftarrow S_i + \bar{a}_i(x_t)$, and a replay *burst* of n_b consecutive batches is triggered when the mean pressure of currently asleep units exceeds θ; consolidation discharges the pressure of the units that received it, $S_i \leftarrow (1 - \mathbf{1}[i \text{ asleep}])\, S_i$. This is a unit-level Process S (Borbély, 1982).

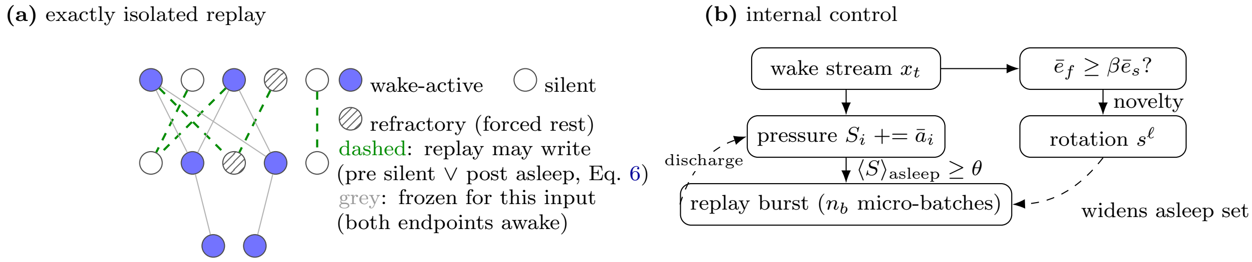

Figure 1: Mechanism. (a) During each waking batch, replay from the episodic buffer is written only into synapses whose update is provably invisible to the current input (Propositions 1–2); refractory rotation forces recently used units into the consolidable set (Corollary 1). (b) Two internal signals: unit-level homeostatic pressure triggers replay bursts; a relative-novelty gate decides when rotation runs.

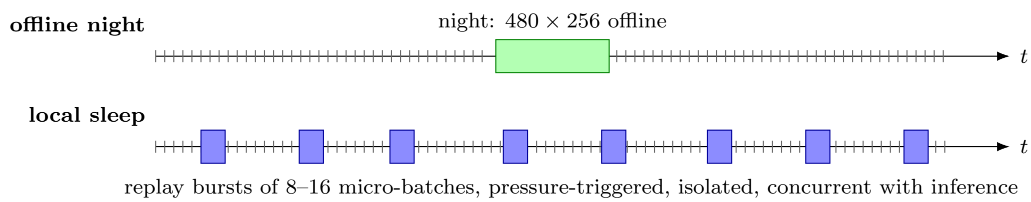

Figure 2: Consolidation schedules. Ticks are waking batches. The offline night pauses the stream; local sleep consolidates during it, in SWR-like bursts confined to asleep units.

Relative novelty (when to rotate). With fast and slow EMAs of the batch error $\bar{e}_f(t) = (1-\alpha_f)\bar{e}_f + \alpha_f e_t$, α_f=0.1, and $\bar{e}_s$, α_s=0.002, rotation runs only while

$$\bar{e}_f(t) \;\geq\; \beta\,\bar{e}_s(t), \qquad \beta \approx 1.1\text{–}1.5, \tag{7}$$

i.e. while the stream is *new relative to the learner's own recent history* (an ACh-like adaptation signal (Hasselmo, 2006)). Monitoring streaming error with exponentially weighted averages is standard in concept-drift detection (Ross et al., 2012), and related loss-ratio triggers appear in concurrent continual-learning work; we claim no priority for the detector itself — only for what it gates (the rotation, hence the replay channel's width) and for the unmastered clause below. Absolute thresholds provably cannot serve here: the converged error of a ten-class i.i.d. stream exceeds any level a two-class task dips under, and we observe exactly this failure (gate open 100% of the time on static MNIST, 95% on CIFAR, for any threshold that behaves on split-MNIST). Because rotation is *beneficial* on streams the learner never masters (Section 5), the final gate adds an "unmastered" clause: with e_0 the mean error over the first 20 batches (a chance-level estimate), rotation runs while

$$\bar{e}_f(t) \;\geq\; \beta\,\bar{e}_s(t) \qquad \text{or} \qquad \bar{e}_f(t) \;\geq\; \gamma\,e_0, \qquad \gamma \approx 0.5. \tag{8}$$

The first clause opens the gate at task switches, the second keeps it open on any stream whose error remains a substantial fraction of chance — both scale-free.

4.4 Cost accounting

Replay cost is measured in samples (R_s = batches × batch size) and synaptic work. The night costs $R_s = 480 \times 256 \approx 1.2 \times 10^5$ samples. Continuous local sleep with micro-batches of 8 costs 18,760 × 8 ≈ 1.5×10^5 samples (1.2× the night); pressure-triggered bursts with batch 64 cost 1.8×10^5 (1.4×).

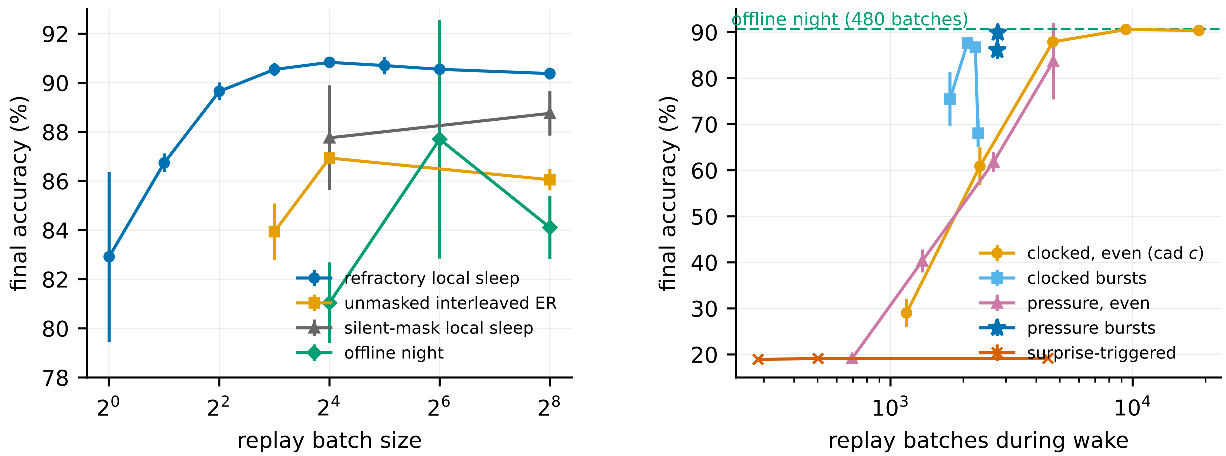

Figure 3: Left (isolation): final split-MNIST accuracy vs. replay batch size. Masked refractory replay is insensitive down to 8 samples; unmasked, silent-masked and offline-night replay all degrade. **Right (timing):** accuracy vs. number of waking replay batches. Even trickles and rare naps fail; bursts of 8–16 succeed; unit-level pressure triggering (stars) dominates at equal budget; surprise triggering fails at any volume.

5 Results

5.1 Local sleep replaces the night

On split-MNIST at K=1000, refractory local sleep with replay every other waking batch (batch 64) reaches $91.3 \pm 0.3\%$ with no offline phase, above the best offline night ($90.7 \pm 0.4\%$) and far above unmasked interleaved ER at the same budget (86.1–86.9%); the stateless-SGD variant of Section 5.3 reaches $\mathbf{92.0} \pm \mathbf{0.6}\%$. Adding a night on top of continuous local sleep yields nothing (90.9 ± 0.3 vs. 90.8 ± 0.1): once consolidation runs during wake, the offline phase is redundant at this buffer size.

5.2 Isolation stabilises micro-batch replay

Figure 3: masked refractory replay is flat in replay batch size from 256 down to 8 samples ($90.4 \to 90.5\%$), while unmasked ER falls to 83.9 ± 1.2, the silent mask to 87.8 ± 2.1, and the offline night to 81.0 ± 1.6 at batch 16. Masking the replay weight update is what makes replay affordable: the earlier "39×" cost of inference-time consolidation was an artefact of counting 256-sample batches.

5.3 The direction of protection matters, and state-free isolation wins

Two ablations sharpen the construction. *Mirror isolation* implements the classical protected direction with our own machinery: the *waking* update is barred from synapses whose both endpoints served the last replay batch (protect the past), while replay itself is applied unmasked. With everything else identical, mirror isolation reaches only $81.3 \pm 2.9\%$ — below even unmasked interleaved ER (86.9 ± 1.2) and far below our direction (90.8 ± 0.1): constraining wake learning impairs the new task while the unmasked replay keeps perturbing the live computation, a double loss. The protected direction is thus not a framing choice but a 9.5-point empirical effect. Second, replacing masked Adam with *stateless* heavy-ball SGD inside the same isolation raises the ceiling: $\mathbf{92.0} \pm \mathbf{0.6}\%$ sequential and $93.3 \pm 0.2\%$ static (six seeds; vs. 90.8/92.0 for masked Adam; unmasked SGD control 84.1 ± 4.1). Carrying no state at all is simpler and better still — and closer to biology; the ablation in Section 6 traces masked Adam's deficit to the confinement of its moments, which the exactness

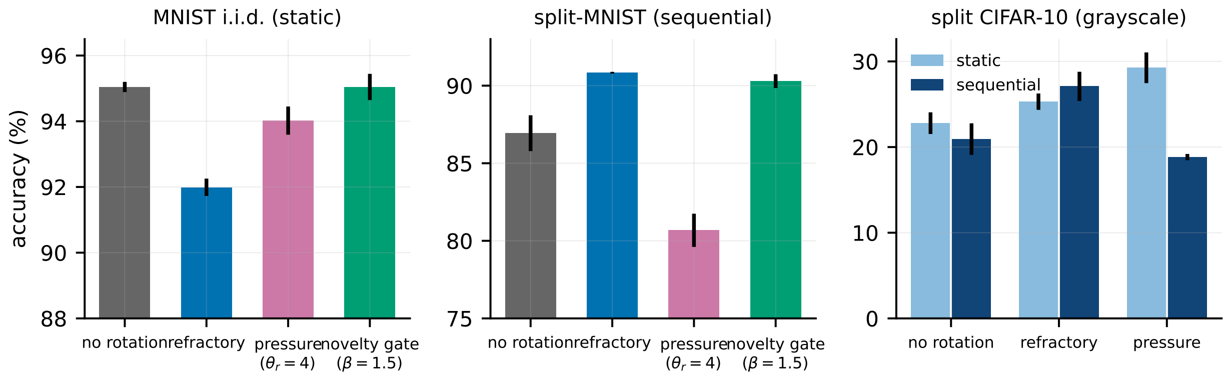

Figure 4: Rotation: price, trade-off, and gate. Left: on i.i.d. MNIST the novelty gate restores no-rotation accuracy. Middle: on split-MNIST rotation is required and the gate keeps it. Right: on split CIFAR-10 (grayscale) rotation is a gain on *both* axes.

guarantee demands but accuracy, it turns out, does not. Third, *soft* rotation (halving, rather than silencing, recent winners) loses on both axes (88.4 ± 0.7 sequential, 86.0 ± 1.3 static): all-or-none OFF periods beat turned-down gain, echoing the biological finding that tonic firing-rate reduction does not discharge local sleep pressure while ON/OFF alternation does (Driessen et al., 2026). Table 1 in Section 6 collects these controls together with the remaining component knockouts and the substrate ladder.

5.4 When replay must be rare: homeostatic bursts, never surprise

Figure 3 (right): at ∼12% of replay events, clocked bursts of 8–16 batches retain 86.8–87.6% while an even trickle collapses (60.9 ± 4.1) and rare long naps also fail (63–70%) — masked replay reaches only ∼half the synapses per event and can neither wait nor dribble. Pressure-triggered bursts reach $\mathbf{89.9 \pm 0.1}$% at 15% of events (2,773 batches; 1.4× night samples with batch 64). Surprise-triggered waking replay fails at *any* volume (≈ 19% even at 4,472 events): novelty fires only after task switches and starves later epochs. The waking trigger must be homeostatic — the opposite of the night's optimal trigger, which our earlier controller showed *should* be novelty-timed.

5.5 The i.i.d. price of rotation and the novelty gate

Figure 4: always-on rotation costs 1.3–3.0 points on i.i.d. MNIST (95.0 → 92.0–93.7). Gating rotation by *unit-level pressure* only trades the axes (static 91.6 → 94.0 as sequential 88.8 → 80.7; no dominant point). The *relative novelty* gate (7) removes the price at no sequential cost on MNIST: static $\mathbf{95.0 \pm 0.4}$ (rotation on 1.4–17% of batches), sequential 90.3 ± 0.4 (β=1.5) or 90.9 ± 0.6 (β=1.1). On split CIFAR-10 the price *reverses*: rotation gains +2.5 (refractory) to +6.5 points (pressure-selected rest) on the static axis — forced rotation acts as a regulariser in the underfit regime — while the pure novelty gate (7) *misfires* there (sequential 19.1 ± 1.1: fast/slow → 1 on a stream that never becomes predictable, so rotation shuts off exactly where it helps). The progress-keyed gate (8) resolves this with one setting across all four regimes: split CIFAR-10 sequential 27.2 ± 0.7 and static 25.3 ± 1.0 (both = always-on rotation), split-MNIST 90.4 ± 0.9, i.i.d. MNIST 94.3 ± 0.2 at γ=0.5 — retaining a residual 0.7-point static cost relative to the pure novelty gate on easy streams, the price of the unmastered clause. (All gate numbers in this subsection use masked Adam; under the stateless optimiser the gate must be committed to minimum-duration bouts — Section 6.)

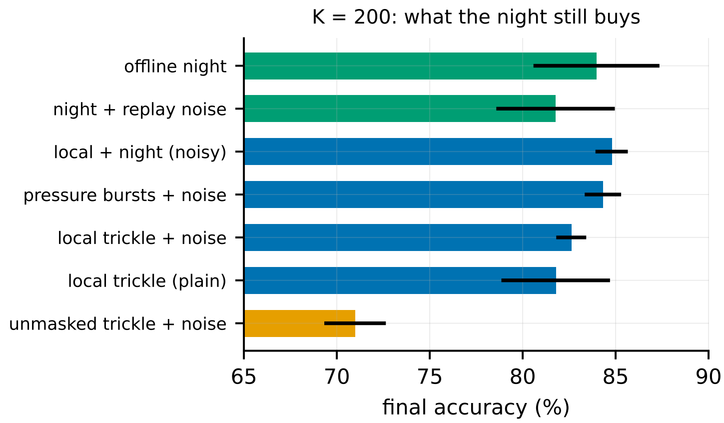

Figure 5: $K = 200$ **mechanism triangulation.** Isolation helps, diversity helps only during wake, and the irreducible advantage of the night at tiny buffers is its interference-free window.

5.6 Transfer to CIFAR-10

On split CIFAR-10 (grayscale, same protocol) the ordering transfers and widens: refractory local sleep $27.1 \pm 1.7 >$ BP+ER $25.1 \pm 1.5 >$ offline night $23.8 \pm 1.2 >$ unmasked ER $20.9 \pm 1.8 >$ no buffer 16.0–16.5, despite task-active code overlap of 0.9–0.98. On the V1-like feature front-end the ordering among local variants persists and widens (refractory $37.0 \pm 0.4 >$ progress-gated $36.0 \pm 0.6 >$ offline night $31.7 \pm 1.5 \approx$ unmasked $30.8 \pm 0.6 \gg$ no buffer 17.0; static rotation gain +3.9: 35.6 ± 1.3 vs. 31.7 ± 3.6), and BP+ER reaches 42.6 ± 0.5 — the reverse of the raw-pixel ordering. In our first report this read as a 5.6-point regime boundary; a two-sided fairness pass revises it. The stateless optimiser transfers to the feature regime (40.6 ± 1.0 sequential at η=0.01, six seeds; static 38.6 ± 1.4), while BP+ER at its own best learning rate gains little (43.3 ± 1.0 at $3 \cdot 10^{-4}$). With both sides tuned the honest gap is 2.7 points, not 5.6: most of the boundary was the local learner's optimiser, not its learning rule. On raw pixels the stateless gain does not appear (27.2 ± 1.4 vs. 27.1 ± 1.7 for masked Adam): statelessness pays in the converged and feature regimes, not in the underfit one. The advantage over *other local schedules* (nights, unmasked ER) remains robust across all three input regimes, and the residual gap to backpropagation-with-replay on features stands as an honest boundary.

5.7 Small buffers: what the night still buys

At K=200 the night appeared to keep a ∼2.6-point edge in our earlier phases; the final triangulation (Figure 5) dissolves most of it into a substrate mismatch. On the *same* 512–256/5% substrate, the night reaches 84.0 ± 3.4 while noisy pressure-burst local sleep reaches 84.3 ± 1.0 and local+night 84.8 ± 0.9 — parity, with three-fold smaller variance for the wake-side scheme. The strongest tiny-buffer number remains the narrower 256–128/10% night (86.9), so the offline window retains an advantage only in combination with narrow substrates. The mechanism results stand: replay diversity (noise σ=0.2) is specifically a wake-side lever — it lifts every local scheme ($81.8 \pm 2.9 \rightarrow 84.3 \pm 1.0$) and *hurts* the night ($84.0 \rightarrow 81.8 \pm 3.2$) — and *unmasked* waking replay collapses (71.0 ± 1.7): isolation is a benefit, not a cost, at every buffer size we tested.

5.8 Energy

Converted to spiking inference (T_s=8), the full system runs at 91.6% and 119 nJ/sample vs. 2461 nJ for dense rate inference (20.7×) and 516 nJ for event-driven rate inference (4.3×); the rotation-trained network loses accuracy beyond T_s=8 (90.4 at 16), unlike its non-rotated control (94.1) — an open interaction between rotation-shaped weights and threshold calibration.

5.9 Sleep-anchored down-selection

Motivated by the synaptic homeostasis hypothesis (Tononi and Cirelli, 2014), we anchored structural plasticity (prune the weakest fraction of live synapses, regrow the same number with a distance bias) to consolidation windows, using the Kolen–Pollack decay of (5) as the natural weakening force on unreplayed synapses. The verdict splits three ways at matched turnover ($\approx$ 5% per epoch). On the *continuous* local-sleep substrate pruning is free (sleep-anchored 90.9 ± 0.2, clocked 90.6 ± 0.4, none 90.8 ± 0.1). On the pure *night* substrate, pruning immediately after each night helps substantially (87.2 ± 3.8 vs. 84.1 ± 1.3) — offline down-selection rescues the wide-sparse night. On the *episodic-burst* substrate pruning is harmful (84.6 ± 0.5 burst-anchored, 86.6 ± 1.3 clocked, vs. 89.9 ± 0.1 without): when replay is rare, synapses are cut before enough consolidation events have protected them. Down-selection, like replay itself, needs either density or an offline window.

5.10 Negative results

For the record, the following were falsified under matched budgets: a frozen dentate-gyrus-style sparse expansion front-end (input decorrelation does not propagate through learned k-WTA layers: task overlap at the input drops $0.79 \to 0.18$ yet every downstream metric worsens by 3–11 points); generative (REM) replay as a substitute for episodic samples; margin-based reverse learning; surprise-gated memory *writing*; multiplicative burst coding; absolute-threshold rotation gates; pruning anchored to sparse replay bursts (above); soft (gain-reduction) rotation and mirror-direction isolation (Section 5.3); an OFF-side gate refractory — the flip-flop's inactive half in this system; strongly coupled synaptic anchors; and utility-exempt rotation as a route past the bout gate (all Section 6).

6 Ablation Study

The system under test is a conjunction of five ingredients: a biologically constrained substrate, the exact isolation mask (6), the refractory rotation, a stateless optimiser, and the progress-keyed gate (8). This section removes or replaces one ingredient at a time, at two levels: the consolidation mechanism (Table 1) and the substrate beneath it (Table 2). Every mechanism cell shares the same substrate (512–256 hidden, 5% k-WTA, 30% distance-dependent connectivity), the same buffer (K=1000, random writing) and the same replay budget (waking batch 16, replay batch 16 after every waking batch), three seeds each.

6.1 Reading the component table

Read top-down, the first block prices each ingredient by what its removal costs the sequential axis under the stateless optimiser: rotation is worth +3.0 points (92.0 vs. 89.0 for the pre-silent channel alone), isolation +7.9 with a six-fold variance reduction (92.0 ± 0.6 vs. 84.1 ± 4.1), and replay is worth everything (19.5 is the forgetting floor). The static column identifies who pays: rotation is the only statically costly component (93.3 always-on against 96.0 for isolation without rotation

Table 1: Component ablation of the consolidation mechanism. Sequential = final accuracy on split-MNIST; static = i.i.d. MNIST (the axis that must not degrade). All rows use the 512–256/5% substrate except the backprop references (dense 256–128). Dashes: combination not applicable. Bold: column best. The bout-gate and always-on rows carry six seeds; all others three.

Variant	Sequential	i.i.d. (static)
Full system and single knockouts (stateless SGD, $\eta = 0.02$)		
full: exact isolation + rotation + per-batch progress gate	89.1 ± 0.3	95.0 ± 0.6
the same gate committed to bouts (min on-duration 2048)	92.0 ± 0.4	94.2 ± 0.4
− gate (rotation always on)	92.0 ± 0.6	93.3 ± 0.2
+ two-timescale anchor (λ=μ=$3{\cdot}10^{-4}$), rotation always on	$\mathbf{92.4 \pm 0.2}$	93.3 ± 0.2
− rotation (pre-silent isolation only)	89.0 ± 0.5	$\mathbf{96.0 \pm 0.1}$
− isolation (unmasked interleaved replay)	84.1 ± 4.1	95.7 ± 0.1
− replay (no episodic buffer; Adam)	19.5 ± 0.0	95.1 ± 0.1
Optimiser state under the mask (rotation always on, Adam)		
moments confined to the mask	90.8 ± 0.1	92.0 ± 0.3
moments advancing outside the mask	91.9 ± 0.4	92.8 ± 0.4
Replacing the construction (Adam)		
offline night instead of concurrent replay, same substrate	84.1 ± 1.3	—
the same night on its preferred narrow substrate	90.7 ± 0.4	—
mirror direction: protect the past (Section 5.3)	81.3 ± 2.9	92.1 ± 0.1
soft rotation: gain 0.5 instead of 0	88.4 ± 0.7	86.0 ± 1.3
External references		
unmasked ER at the same budget (Adam)	86.9 ± 1.2	94.8 ± 0.2
backprop (+ER sequential; plain static), dense	89.3 ± 0.8	97.5 ± 0.0

— the highest static cell in the table, above even the no-buffer baseline), and the progress gate refunds the price in full (95.0, at the 95.1 ceiling). The *per-batch* gated conjunction, however, is the best system nowhere: against always-on rotation it loses 2.9 sequential points (89.1 vs. 92.0), and against the rotation-free system it is sequentially identical (89.1 vs. 89.0) yet a full static point worse (95.0 ± 0.6 vs. 96.0 ± 0.1) — it manufactures no operating point that always-on rotation and plain silent isolation do not already offer. Under Adam the same per-batch gate *is* genuinely adaptive (90.4 gated vs. 87.8 silent sequentially, at a modest static cost, 94.3 vs. 95.0 ± 0.3), so the failure is specific to the stateless optimiser, whose rotation gain the gate's flicker destroys: the trigger is open on only 12% of batches, and within each task it closes the moment the task is mastered — precisely the batches on which continuous rotation earns its +3.0.

Committing the gate to *episodes* repairs it. In the bout-committed variant, every firing of the same trigger re-arms a rotation episode with a minimum on-duration of M waking batches — the consolidated-bout structure of the sleep–wake flip-flop switch (Saper et al., 2005), in which state flicker is the pathology, not the mechanism. The sparse trigger is thereby converted into long contiguous episodes whose coverage is regime-asymmetric: at M=2048, bouts chain into near-permanent rotation on the sequential stream (94% of batches) yet cover only 63% of the static one. The result — 92.0 ± 0.4 sequential at 94.2 ± 0.4 static over six seeds — matches always-on rotation exactly on the sequential axis (92.0 ± 0.6) while refunding 0.9 of its static price (93.3 ± 0.2): a clean weak domination that retires always-on rotation as an operating point. Bout length is a proper knob: M=512 (duties 0.61/0.38) gives 91.7 ± 0.4/94.0 ± 0.5, already recovering most of what the per-batch gate loses, while M=4096 drives the static duty to 0.94 and the refund vanishes (93.4 ± 0.1). The static ceiling of silent isolation (96.0) stays out of reach — rotation, even in bouts, keeps a residual price — so the committed gate buys regime-adaptivity, not a free lunch. Completing the flip-flop with an OFF-side refractory (once a bout expires, triggers are ignored for M_{off} batches) is a clean null: short refractories ($M_{\text{off}} \leq 1024$) never engage on the static stream — a bout expires only after its trigger has been quiet for M batches, so static re-triggering is not flicker to begin with — while a long one (M_{off}=4096) lowers the static duty to 0.53 without moving static accuracy (94.0 ± 0.7) and monotonically erodes the sequential axis ($92.1 \rightarrow 91.0$, occasionally masking a task switch). Together with the M=512 cell (duty 0.38, static 94.0), this locates the residual static price in the rotation of settled coalitions itself, not in how often the gate fires.

The middle block tests the exactness requirement itself and yields the sharpest surprise of the table. Propositions 1–2 require Adam's moments to advance only inside the mask: state advancing outside re-injects the replay gradient into awake synapses through the next unmasked waking step, and output-neutrality is lost. Empirically, however, the leak *helps* — 91.9 ± 0.4 vs. 90.8 ± 0.1 sequential, 92.8 ± 0.4 vs. 92.0 ± 0.3 static. Exact state confinement, not isolation, is what held masked Adam below the stateless optimiser: moments frozen outside each batch's mask go stale across a unit's on/off cycles (Bricken et al., 2023), and either remedy — letting the state track every gradient (surrendering the guarantee) or carrying no state at all — recovers the same ≈ 1 point. Statelessness is the only choice that keeps the guarantee and the accuracy at once: the exactness of Propositions 1–2 costs about one point when bought from Adam, and nothing when bought from heavy-ball SGD.

Two further notes complete the picture. The stateless gain also vanishes at half cadence: SGD with replay batch 64 every other waking batch reaches 91.3 ± 0.3, identical to masked Adam on the same schedule — statelessness pays only while consolidation runs continuously. And the replacement block collects, in one place, that every structural alternative loses on the sequential axis: the offline night on this substrate (84.1), the protected-past direction (81.3), softened rotation (88.4), unmasked ER (86.9), with the strongest competitor — the night on its preferred narrow substrate (90.7) — still below the always-on operating point.

6.2 Sensitivity to the learning rate

The stateless comparison invites an objection: SGD's η was chosen, Adam's was left at its default. A two-sided sweep dissolves it. Under always-on rotation, SGD holds a sequential plateau over $\eta \in [0.01, 0.02]$ (91.9 ± 0.5/92.0 ± 0.6) and a static plateau over $[0.02, 0.05]$ (93.3–93.7), decaying gracefully outside (86.3/89.2 at η=0.1); η=0.02 is the joint optimum. Masked Adam, swept over a tenfold range, has its sequential optimum exactly at the default (90.8 ± 0.1 at 10^{-3}; 88.6 at $3 \cdot 10^{-4}$, 88.9 at $3 \cdot 10^{-3}$) — the comparison of Section 5.3 was already at Adam's best — and no Adam setting reaches the stateless joint point ($3 \cdot 10^{-3}$ buys 93.6 static at a three-point sequential cost). Unmasked SGD is poor and unstable at *every* rate (84–88, seed deviations 1.8–4.1, three- to eight-fold those of the masked runs), making the stabilising role of isolation a cross-η fact. The bout-committed gate inherits the plateau, and the optimum shifts one notch down on CIFAR (η=0.01).

6.3 Two probes of the residual static price

The OFF-refractory null located the bout gate's residual static price (94.2 vs. 96.0 rotation-free) in the rotation of settled coalitions itself. Two mechanisms attack that diagnosis directly; one is a clean negative, one a modest positive.

Utility-exempt rotation spares the top-q units by long-term use trace from ever being benched, so the settled coalitions stay awake. It traces a smooth trade-off between always-on rotation and the rotation-free system (q=0.10: 90.9 ± 0.1/94.2 ± 0.2; q=0.25: 89.1 ± 0.4/95.4 ± 0.2), but the bout

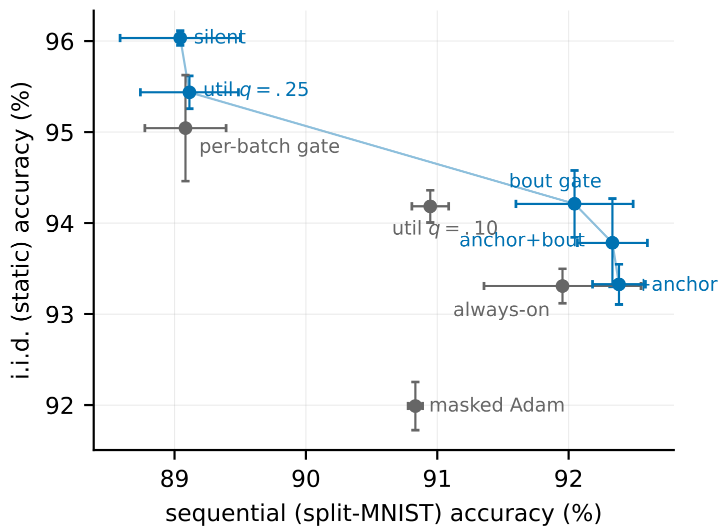

Figure 6: The operating frontier under the stateless optimiser: split-MNIST sequential accuracy against i.i.d. static accuracy (six seeds on the bout-gate and always-on points, three elsewhere). Blue: non-dominated configurations; grey: dominated ones. No single configuration wins both axes; the bout-committed gate is the sequential-side default, silent isolation the static-side one.

gate beats the q=0.10 point by +1.1 sequential at equal static: *temporal commitment dominates structural exemption*, recapitulating at the utility level what unit-level pressure gating showed earlier (Figure 4) — any structural narrowing of the rotation pool is paid for in replay-channel width.

Two-timescale synapses give every weight a slow anchor in the spirit of synaptic consolidation cascades (Benna and Fusi, 2016): the fast weight decays toward its anchor at rate λ while the anchor absorbs it at rate μ, ticking only on waking updates so the exactness of the replay path is untouched. The strong hypothesis — that settled function living in the anchor would dissolve rotation's static price — is falsified: across $\lambda \in [3{\cdot}10^{-4}, 3{\cdot}10^{-3}]$, $\mu \in [10^{-4}, 10^{-3}]$ no cell exceeds the anchor-free static accuracy, and strong coupling hurts both axes (the anchor becomes a lagging drag). Weak symmetric coupling (λ=μ=$3{\cdot}10^{-4}$), however, acts as a *sequential stabiliser*: 92.4 ± 0.2, the best sequential cell in Table 1 with the seed variance halved, at static parity — weakly dominating plain always-on rotation. Anchor and bout gate compose sub-additively ($92.3 \pm 0.3/93.8 \pm 0.5$, between the parents). Figure 6 draws the resulting operating frontier.

6.4 The price of the substrate

Table 2 traces the path from a dense backpropagation MLP to the substrate used throughout, on i.i.d. MNIST where each constraint's cost is cleanest. The upper block is cumulative; the lower block replaces one ingredient at its stage and is to be compared with the matching cumulative row.

The cumulative ladder prices the entire biological constraint set at 1.2 points below the dense backpropagation reference (97.7 $\to$ 96.5) for $4\times$ fewer synaptic operations per pass; the local burst-error objective itself costs nothing (97.8 unconstrained). No single constraint is expensive: the bounded-error/Dale/no-transport trio costs 0.7, 10% k-WTA one tenth of a point, and the 70% connectivity cut half a point — of which distance-dependent wiring recovers +1.1 over random wiring at equal density. The lower block adds three cautions. Settling cannot merely be shortened (T=1 collapses to 17.9); the single-phase burst sweep is the principled replacement, matching the T=5 accuracy in a single pass. The Kolen–Pollack decay and even fully restored weight transport

Table 2: Substrate ablation on i.i.d. MNIST (256–128 hidden core, three seeds). "Syn. ops" is the fraction of the dense MLP's synaptic operations in one forward pass. Upper block: constraints added cumulatively; lower block: one ingredient replaced at its stage.

Substrate	Accuracy	Syn. ops
dense backpropagation MLP (autograd reference)	97.7 ± 0.1	1.00
local burst-error learner, unconstrained	97.8 ± 0.1	0.85
+ bounded error, Dale's law, sign-concordant feedback (no transport)	97.1 ± 0.0	0.85
+ k-WTA 10%	97.0 ± 0.2	0.80
+ relaxation T=5	97.0 ± 0.1	0.80
+ 30% distance-dependent connectivity	96.5 ± 0.2	0.24
+ burst gate and burst baseline	96.3 ± 0.2	0.24
+ single-phase burst sweep replacing relaxation (*final substrate*)	96.5 ± 0.1	0.24
random wiring instead of distance-dependent, equal density	95.4 ± 0.0	0.24
one settling step (T=1) instead of five	17.9 ± 6.8	0.80
weight transport restored (mirroring, no KP decay)	96.3 ± 0.2	0.24
Kolen–Pollack decay removed	96.3 ± 0.1	0.24
heavy-ball SGD instead of Adam ($\eta = 0.02$)	95.2 ± 0.3	0.24
k-WTA 5% / 2% instead of 10%	95.9 ± 0.1 / 94.2 ± 0.2	0.79/0.78
multiplicative burst coding	78.8 ± 1.9	0.24
training on T_s=8 spike counts	80.0 ± 0.8	0.24

are accuracy-neutral at this scale — the decay earns its keep as a regulariser on harder inputs, not as an aligner. And the two spiking-flavoured training schemes fail outright: spiking is affordable at *inference* (the energy measurements above), not as a training-time code.

Finally, the two tables disagree about the optimiser in an instructive way: on the plain substrate SGD *loses* 1.3 points to Adam (95.2 vs. 96.5), yet inside the isolated-consolidation loop it wins on both axes. Per-synapse adaptive state is an asset while every synapse trains on every batch and a liability once updates become intermittently mask-confined — the staleness mechanism of Table 1's middle block seen from the other side. The optimiser is not a neutral implementation detail: it interacts with the sleep machinery, and the right choice reverses between the static substrate and the consolidating system.

7 Discussion

Why isolation stabilises micro-batches. A replay micro-batch is a high-variance gradient estimate. Applied to the full network it perturbs the very coalition currently encoding the wake stream, and with shared optimiser state the perturbation compounds (cf. the unmasked curve of Figure 3). Inside the mask, each update is provably output-neutral for the present input (Propositions 1–2), so variance accumulates only where it can do no immediate harm and is averaged over thousands of events. The offline night enjoys no such protection: with nothing clamping the network, a small noisy batch at 3× learning rate moves the whole weight state.

Correspondence with biology. We treat the biology as motivating analogy, not as a claim of mechanistic equivalence. The refractory rule mirrors the causal finding of Driessen et al. (2026) that ON/OFF *alternation*, not tonic silencing, discharges local sleep pressure and rescues memory — our silent-mask control (their tonic condition) is reliably worse. The dissociation of triggers — homeostatic for waking consolidation, novelty-timed for full nights — is a testable prediction for biology. The regime dependence of the rotation price (cost on easy streams, gain under chronic underfitting) may bear on why local sleep in vivo both impairs behaviour (Vyazovskiy et al., 2011) and serves useful functions when appropriately timed.

Honest limits. All results are at MLP scale on MNIST/CIFAR variants with three seeds (six on the headline rows); BP+ER keeps a 2.7-point lead on feature-based CIFAR with both sides at their best learning rates, bounding the regime of our headline comparison; the progress gate keeps a 0.7-point static residue under Adam, must be committed to minimum-duration bouts under the stateless optimiser, and even then leaves a 1.8-point static gap to rotation-free isolation (Section 6); and the novelty claims rest on our own literature search (August 2026); individual primitives — refractory competition, sparse or null-space isolation, replay, adaptive scheduling — all have ancestors, and the claimed contribution is their combination into concurrent, exactly isolated consolidation.

8 Future Work

(i) Explaining why backpropagation with replay overtakes the local learner specifically on feature representations (dense connectivity, per-synapse adaptivity, or the learning rule itself); (ii) the bout gate's residual static price (94.2 vs. 96.0 rotation-free): the OFF-side refractory, utility exemption and synaptic anchoring are now all ruled out as routes past it (Section 6), leaving mastery-annealed rotation depth untried; (iii) the narrow-vs-wide substrate interaction of offline nights at tiny buffers; (iv) convolutional and deeper stacks; (v) a throughput theory of the isolated channel predicting the required replay cadence from ρ_ℓ, code overlap and the interference rate; (vi) the rotation–spiking calibration interaction beyond T_s=8.

9 Conclusion

Consolidation does not require an offline phase. With exactly isolated replay, refractory rotation of the active coalition, homeostatic burst triggering and a relative-novelty gate on the rotation itself, a biologically constrained learner consolidates *while it infers*: it matches and slightly exceeds the strongest offline-night schedule on split-MNIST at $\sim$1.2$\times$ the night's replay samples, pays nothing on learnable i.i.d. streams, gains outright on harder data, and runs at a twentieth of dense-rate inference energy when spiking. The offline night retains one irreducible role — consolidating very small episodic stores — which sharpens, rather than blurs, the division of labour between wake and sleep.

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